Linear Programming

Problem 1

A baker makes two types of treats: **Cupcakes (x)** and **Cookies (y)**.

- **Time Constraint:** It takes 2 hours to bake a batch of cupcakes and 1 hour for cookies. He has **8 hours** available.
- **Space Constraint:** He can only fit **5 total batches** in his display case.
- **Objective:** He makes a profit of **$30** per batch of cupcakes and **$20** per batch of cookies. Find the number of cupcake batches (x) and cookie batches (y) that maximize the profit.

optimization problem under some constraints.

(*) Time constraint:

Time needed to make x batches of cupcakes : $2x$

Time needed to make y batches of cookies : y

In total, time needed to make x batches of cupcakes and cookies :

$2x + y$ (hours)

$$\Rightarrow \boxed{2x + y \leq 8}$$

(*) Space Constraint :

$$x + y \leq 5$$

(*) Other constraints:

$x \geq 0$, $y \geq 0$

<u>Objective</u>: Maximize Profit $= 30x + 20y$

<u>Step 1</u>: List all the lines (Linear Equations)

(A) $2x + y = 8$

(B) $x + y = 5$

(C) $x = 0$

(D) $y = 0$

<u>Step 2</u>: Find All possible intersections of the lines

<u>(A) and (B)</u>:

$$\begin{array}{rl} 2x + y = 8 & \text{(A)} \\ - \quad x + y = 5 & \text{(B)} \\ \hline x = 3 & \end{array}$$

$x + y = 5 \Rightarrow 3 + y = 5$

$\Rightarrow y = 2$

$\Rightarrow (x, y) = (3, 2)$

<u>(A) and (C)</u> :

$2x + y = 8$ (A)

$x = 0$ (C)

$\Rightarrow 2 \times 0 + y = 8 \Rightarrow y = 8$

$\Rightarrow (x, y) = (0, 8)$

<u>(A) and (D)</u>

$2x + y = 8$ (A)

$y = 0$ (D)

$\Rightarrow (x, y) = (4, 0)$

<u>(B) and (C)</u> : $x + y = 5$ (B)

$x = 0$

$\Rightarrow (0, 5)$

(B) and (D) :

$x + y = 5$ (B)

$y = 0$ (D)

$\Rightarrow (x, y) = (5, 0)$

(C) and (D) :

$x = 0$ (C)

$y = 0$ (D)

$\Rightarrow (x, y) = (0, 0)$

<u>Step 3 and Step 4</u> : check for vertices and the maximum

x	y	$2x + y \leq 8$	$x + y \leq 5$	Profit $= 30x + 20y$
3	2	Yes	Yes	130
0	8	Yes	NO	
4	0	Yes	Yes	120
0	5	Yes	Yes	100
5	0	NO	Yes	
0	0	Yes	Yes	0

Solution $x = 3$, $y = 2$

The Math: Problem 1 (Maximization)

Variables:

- x = Number of batches of Cupcakes
- y = Number of batches of Cookies

Objective Function:

Maximize the total profit (P):

$$P = 30x + 20y$$

Subject to Constraints:

1. **Time:** $2x + y \leq 8$ (Each cupcake batch takes 2 hours; cookies take 1)
2. **Space:** $x + y \leq 5$ (Total batches cannot exceed 5)
3. **Non-negativity:** $x \geq 0, y \geq 0$

You try

A small shop builds two types of custom computers: Gaming PCs (x) and Office PCs (y).

- **Time Constraint**: A Gaming PC takes 3 hours to build, and an Office PC takes 1 hour. The shop has 9 hours of building time available today.
- **Component Constraint**: Due to a parts shortage, they can only build a total of 5 computers maximum.
- **Objective**: They earn $50 for every Gaming PC and $30 for every Office PC. Find the number of Gaming PCs (x) and Office PCs (y) that maximize the profit.

Problem 2

An athlete needs to meet daily nutritional requirements using **Protein Shakes (x)** and **Energy Bars (y)**.

x

y

- **Protein Constraint:** Each shake has 2 units of protein, and each bar has 1 unit. He needs at least **5 units** a day.
- **Energy Constraint:** Each shake has 1 unit of energy, and each bar has 3 units. He needs at least **10 units** a day.
- **Objective:** Each shake costs **$4** and each bar costs **$3**. Find the number of Protein Shakes (x) and Energy Bars (y) that minimize the cost.

* Protein constraint :

$$2x + y \geq 5$$

* Energy constraint

$$x + 3y \geq 10$$

* Other constraints

$$x \geq 0 \quad ; \quad y \geq 0$$

* objectives: minimize cost $= 4x + 3y$

Step 1 : List all the lines (Linear Equations)

(A) $2x + y = 5$

(B) $x + 3y = 10$

(C) $x = 0$

(D) $y = 0$

Step 2 : Find ALL possible Intersections between the lines.

(A) and (B)

$$2x + y = 5$$
$$x + 3y = 10$$

multiply 3 to both sides.

$$\begin{array}{rl} & 6x + 3y = 15 \\ - & x + 3y = 10 \\ \hline & 5x = 5 \end{array}$$

$$\Rightarrow x = 1$$

plug $x = 1$ into $2x + y = 5$

$$\Rightarrow 2 + y = 5$$

$$\Rightarrow y = 3$$

$\Rightarrow \quad (x, -1) = (1, 3)$

(A) and (C)

$2x + y = 5 \qquad (A)$

$x = 0 \qquad (C)$

$\Rightarrow \quad (x, y) = (0, 5)$

(A) and (D)

$2x + y = 5 \qquad (A)$

$y = 0 \qquad (D)$

$(x, y) = (2.5, 0)$

(B) and (C)

$x + 3y = 10$

$x = 0, \quad y = 10/3$

(B) and (D)

$x + 3y = 10$

$y = 0$

$\Rightarrow (x, y) = (10, 0)$

(C) and (D)

$(x, y) = (0,0)$

Step 3 and Step 4 : check for vertices and minimum

x	y	$2x + y \geq 5$	$x + 3y \geq 10$	cost $= 4x + 3y$
1	3	✓	✓	13
0	5	✓	✓	15
2.5	0	✓	NO	
0	10/3	NO		
10	0	✓	✓	40
0	0	NO	NO	

You try

A landscaping company needs to spread mulch and stone using two sizes of trucks: **Small Trucks (x)** and **Large Trucks (y)**.

- **Mulch Capacity Constraint:** A Small Truck carries 1 ton, and a Large Truck carries 3 tons. They need to deliver at least **6 tons** of mulch.
- **Stone Capacity Constraint:** A Small Truck carries 2 tons, and a Large Truck carries 1 ton. They need to deliver at least **7 tons** of stone.

- **Objective:** Running a Small Truck costs **$100** and a Large Truck costs **$200**. Find the number of Small Trucks (x) and Large Trucks (y) that minimize the cost.